

I like what we did the correct the flow of the first part of the app. However, that leaves a lot in the back part of the app. With that, I think the back part is very busy and too much for one section. We need to break it up. So, conceptually, I thought this through and here is where I am. The flow now is:

Set up: title your project and describe your study → upload data set and data structure → confirm your study → analysis – report

but I think, the back-end, is too much: it includes both qualitative and quantitative both in one area plus a lot of meaningful mix-methods techniques.

This should be broken into parts, like we did with the front end. The back should really be 3 of 4 parts. Here is what the steps should be:

Set Up → Structure Data → Choose Design → Analyze in the Right Order → Integrate Evidence → Defend the Report

So, after our second part, confirm your study, we should lend more into the framework of Mixed-Methods. Which follows the pattern of three types of approaches: Explanatory, Exploratory, and Convergent. So, after confirm your study step 3, the user is asked what type of study are wanting to do: they pick one of the three pathways: From there the flow is:

| Pathway A: Explanatory                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Pathway B: Exploratory                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Pathway C: Convergent                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>1st: quantitative Analysis:</p> <ul style="list-style-type: none"> <li>• Predictor/outcome assignment</li> <li>• Suggested tests</li> <li>• Correlation, t-test, chi-square, ANOVA</li> <li>• Effect sizes</li> <li>• Sample sizes</li> </ul> <p>2nd: Qualitative Explanation</p> <ul style="list-style-type: none"> <li>• Builder</li> <li>• Theme discovery</li> <li>• Sentiment</li> <li>• Category editing</li> </ul> <p>3rd. Integration</p> <ul style="list-style-type: none"> <li>• Joint Display</li> <li>• Integration paragraphs</li> <li>• Alignment tab</li> </ul> | <p>1<sup>st</sup>. Qualitative Exploration</p> <ul style="list-style-type: none"> <li>• Quality Brief</li> <li>• Builder</li> <li>• Auto/guided/hybrid/user modes</li> <li>• Per-question themes</li> <li>• Sentiment</li> <li>• Category editing</li> </ul> <p>2<sup>nd</sup> Variable Builder</p> <ul style="list-style-type: none"> <li>• Theme presence</li> <li>• Theme intensity</li> <li>• Sentiment label</li> <li>• Sentiment numeric</li> <li>• Response length</li> </ul> <p>3rd Quantitative Testing</p> <ul style="list-style-type: none"> <li>• Suggested tests</li> <li>• Predictor/outcome roles</li> </ul> | <p>1<sup>st</sup>. Qualitative Track</p> <ul style="list-style-type: none"> <li>• Quality Brief</li> <li>• Builder</li> <li>• Categories</li> <li>• Sentiment</li> <li>• Representative quotes</li> </ul> <p>2nd. Quantitative Track</p> <ul style="list-style-type: none"> <li>• Dataset</li> <li>• Analysis recommendations</li> <li>• Statistical tests</li> </ul> <p>3rd. Compare Evidence</p> <ul style="list-style-type: none"> <li>• Joint Display</li> <li>• Score-to-Theme</li> </ul> |

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| <ul style="list-style-type: none"> <li>• Strength Check</li> <li>• Report</li> </ul> | <ul style="list-style-type: none"> <li>• Export to CSV, Excel, SPSS</li> </ul> <p>4<sup>th</sup>. Integration</p> <ul style="list-style-type: none"> <li>• Joint Display</li> <li>• Integration paragraphs</li> <li>• Strength Check</li> <li>• Report</li> <li>• Claim support audit</li> <li>• Limitations generator</li> </ul> | <ul style="list-style-type: none"> <li>• Alignment</li> </ul> <p>4th. Integration</p> <p>Current pieces:</p> <ul style="list-style-type: none"> <li>• Integration paragraphs</li> <li>• Strength Check</li> <li>• Report</li> </ul> <p>Missing pieces to add:</p> <ul style="list-style-type: none"> <li>• Meta-inference generator</li> <li>• Reviewer-readiness score</li> <li>• Claims audit</li> <li>• Limitations section</li> </ul> |
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## 1. Title Project

Name the project and describe the study.

## 2. Upload / Connect Data

Upload data, paste responses, or connect an existing ReliCheck survey.

## 3. Confirm Data Structure

Identify:

- open-ended responses
- closed-ended variables
- demographics/groups
- outcomes
- time/wave variables

## 4. Choose Mixed-Methods Design

The user selects:

Explanatory

Quantitative first. Use qualitative data to explain the numbers.

Exploratory

Qualitative first. Use themes to build variables, scales, or later quantitative analysis.

Convergent

Qualitative and quantitative analyzed separately, then compared and integrated.

This is the pivot point.

Then the workflow should branch

If Explanatory

Start with:

Quantitative Analysis → Qualitative Explanation → Joint Display → Integration → Report

Use when the user already has scores, group differences, regression results, or survey findings and wants comments to explain them.

If Exploratory

Start with:

Explore Themes → Build Codebook → Create Variables → Quantitative Analysis → Integration → Report

Use when the open-ended data is the starting point and the user wants to turn patterns into measurable variables.

If Convergent

Run parallel tracks:

Qualitative Analysis + Quantitative Analysis → Alignment Check → Joint Display → Integration → Report

For ReliCheck, “Analysis” should become three separate stages:

1. Explore Themes
2. Analyze Numbers
3. Integrate Evidence

That is much clearer for users and more faithful to mixed-methods methodology.

Best structure

Use this:

Strengthening The Platform (thinks to keep in mind)

ReliCheck Mixed-Methods Studio Flow

### 1. Project Setup

Current pieces:

- Title project
- Describe study
- Save/resume project

Missing pieces to add:

- Study purpose prompt
- Intended audience: dissertation, evaluation, HR, accreditation, market research
- Final report type selection

### 2. Data Intake

Current pieces:

- Upload file
- Paste responses
- Link ReliCheck survey
- No data yet option

Missing pieces to add:

- Data privacy notice
- Dataset readiness warning
- Minimum data requirements by pathway

### 3. Confirm Data Structure

Current pieces:

- Response ID
- Open-ended columns

- Closed-ended variables
- Demographics/group variables
- Outcome variables
- Time/wave variables

Missing pieces to add:

- “What can this dataset answer?” checker
- Missing variable warning
- Multi-question open-ended detection
- Automatic design-fit warning

#### 4. Choose Mixed-Methods Design

This should become the major pivot.

Options:

- Explanatory: Quant  $\rightarrow$  Qual  $\rightarrow$  Integration
- Exploratory: Qual  $\rightarrow$  Quant  $\rightarrow$  Integration
- Convergent: Qual + Quant  $\rightarrow$  Integration

Missing pieces to add:

- Design justification statement
- Design map
- Suggested pathway preview
- “What this design can and cannot claim” warning

#### Pathway A: Explanatory

##### 5A. Quantitative Analysis

Current pieces:

- Predictor/outcome assignment
- Suggested tests
- Correlation, t-test, chi-square, ANOVA
- Effect sizes
- Sample sizes

Missing pieces to add:

- Regression
- Logistic regression
- Group comparison assistant
- Assumption checks
- Multiple comparison warning
- Plain-language interpretation

## 6A. Qualitative Explanation

Current pieces:

- Builder
- Theme discovery
- Sentiment
- Category editing

Missing pieces to add:

- Codebook builder
- Inclusion/exclusion criteria
- Representative quote selector
- Theme explanation prompts tied to quant results

## 7A. Integration

Current pieces:

- Joint Display
- Integration paragraphs
- Alignment tab
- Strength Check
- Report

Missing pieces to add:

- Claim support audit
- Contradiction finder
- Limitations generator
- Reviewer-readiness score

## Pathway B: Exploratory

### 5B. Qualitative Exploration

Current pieces:

- Quality Brief
- Builder
- Auto/guided/hybrid/user modes
- Per-question themes
- Sentiment
- Category editing

Missing pieces to add:

- Formal codebook
- Grounded theory-lite option
- Thematic analysis pathway
- Framework analysis pathway
- Intercoder agreement
- Memo writing
- Theme saturation check

## 6B. Variable Builder

Current pieces:

- Theme presence
- Theme intensity
- Sentiment label
- Sentiment numeric
- Response length

Missing pieces to add:

- Construct builder
- Scale builder
- Theme clustering
- Code-to-variable audit
- Variable naming assistant

## 7B. Quantitative Testing

Current pieces:

- Suggested tests
- Predictor/outcome roles
- Export to CSV, Excel, SPSS

Missing pieces to add:

- Reliability analysis for created scales

- Factor analysis
- Regression
- Cluster comparison
- Validation warnings

## 8B. Integration

Same as above:

- Joint Display
- Integration paragraphs
- Strength Check
- Report
- Claim support audit
- Limitations generator

## Pathway C: Convergent

### 5C. Qualitative Track

Current pieces:

- Quality Brief
- Builder
- Categories
- Sentiment
- Representative quotes

Missing pieces to add:

- Codebook
- Intercoder review
- Theme saturation
- Qualitative memo trail

### 6C. Quantitative Track

Current pieces:

- Dataset
- Analysis recommendations
- Statistical tests

Missing pieces to add:

- Descriptive dashboards



- Regression
- Group comparisons
- Reliability
- Assumption checks

## 7C. Compare Evidence

Current pieces:

- Joint Display
- Score-to-Theme
- Alignment

Missing pieces to add:

- Convergence/divergence table
- Contradiction analysis
- Mixed evidence flags
- “Qual supports / complicates / contradicts quant” labels

## 8C. Integration

Current pieces:

- Integration paragraphs
- Strength Check
- Report

Missing pieces to add:

- Meta-inference generator
- Reviewer-readiness score
- Claims audit
- Limitations section

## Universal Final Stage: Report and Defend

This should appear for all three pathways.

Current pieces:

- Editable report
- Version history
- Notes
- AI rewrite
- DOCX/PDF export

- Strength Check appendix

Missing pieces to add:

- Methods appendix generator
- Design justification paragraph
- Limitations section
- Audit trail export
- Reviewer questions checklist
- Dissertation/evaluation/accreditation/HR report templates

Cleanest product language

The full studio should become:

## Most Common Mixed-Methods Pairings

Here are the combinations most often seen in real studies:

| Quantitative        | Qualitative        | Purpose                         |
|---------------------|--------------------|---------------------------------|
| Descriptive stats   | Thematic analysis  | Explain survey patterns         |
| Regression          | Interviews         | Explain predictors/mechanisms   |
| Factor analysis     | Focus groups       | Validate constructs             |
| Cross-tabs          | Open coding        | Understand subgroup differences |
| Longitudinal trends | Narrative analysis | Explain change over time        |
| Cluster analysis    | Case studies       | Interpret hidden group patterns |

ReliCheck is already pointed in the right direction. Its best feature is not the Builder, the Dataset tab, or even the Report module by itself.

Its real advantage is the **integration workflow**.

To lead the industry, ReliCheck should own this promise:

**From open-ended responses to coded themes, from coded themes to variables, from variables to analysis, from analysis to joint display, and from joint display to a defensible mixed-methods report.**

## ow ReliCheck becomes an industry leader

ReliCheck can lead this space if it positions itself as:

**The mixed-methods credibility platform, not just an AI coding tool.**

To become an industry leader, build around five signature capabilities:

### 1. The ReliCheck Mixed-Methods Strength Index

A score that evaluates whether the project is methodologically defensible.

It could include:

- Data readiness
- Theme coverage
- Codebook quality
- Quant/qual alignment
- Test appropriateness
- Integration strength
- Report defensibility

### 2. Reviewer-Ready Audit Trail

Every claim should trace back to:

- Theme
- Quote
- Respondent row
- Variable
- Statistical result
- Report paragraph

This is already hinted at in the guide's citation-chain concept. Make it central.

### 3. Design-Aware Analysis

Do not just recommend tests. Recommend analyses based on the selected mixed-methods design.

For example:

- Explanatory sequential
- Exploratory sequential
- Convergent parallel
- Embedded design

- Evaluation design

#### 4. Human-in-the-Loop Qualitative Control

AI can recommend, but the researcher must approve, revise, merge, reject, and document.

This is already part of the premise: “The studio recommends. You accept or adjust.”

Make that a product philosophy.

#### 5. Publication / Dissertation / Evaluation Report Modes

Let users choose the output standard:

- Dissertation chapter
- Program evaluation report
- Accreditation evidence report
- HR insights report
- Market research report
- Academic manuscript findings section

#### Bottom line

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**From open-ended responses to coded themes, from coded themes to variables, from variables to analysis, from analysis to joint display, and from joint display to a defensible mixed-methods report.**